

Boss Challenge — Paper 10 (Class 7)

Symmetry | Light | Arithmetic Expressions | Physical & Chemical Changes

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A figure has a line of symmetry when a mirror placed on that line shows that:
(A) the two halves are different shapes
(B) the figure has no halves at all
(C) one half is the exact mirror image of the other half
(D) the figure becomes larger
2. In a uniform medium like clear air, light travels:
(A) in zig-zag steps
(B) in curved paths
(C) in widening spirals
(D) in straight lines
3. In the expression $3 + 4 \times 2$, which operation must be carried out first?
(A) whichever is written on the left
(B) the multiplication (4×2)
(C) both at the same time
(D) the addition ($3 + 4$)
4. A change in which NO new substance is formed is called a:
(A) chemical change
(B) nuclear change
(C) biological change
(D) physical change
5. How many lines of symmetry does a square have?
(A) 1
(B) 2
(C) 8
(D) 4
6. We are able to see an ordinary (non-luminous) object such as a book because:
(A) the book absorbs all the light
(B) our eyes send out rays to it
(C) the book makes its own light
(D) light from a source bounces off it into our eyes
7. Evaluate $3 + 4 \times 2$.
(A) 14
(B) 24
(C) 11
(D) 9

8. A change in which one or more NEW substances are formed is called a:
- (A) physical change
 - (B) temporary change
 - (C) chemical change
 - (D) seasonal change
9. How many lines of symmetry does a rectangle (that is not a square) have?
- (A) 2
 - (B) 0
 - (C) 1
 - (D) 4
10. The bouncing back of light when it strikes a shiny surface is called:
- (A) absorption
 - (B) germination
 - (C) dispersion
 - (D) reflection
11. First add inside the brackets, then work out the value of $(3 + 4) \times 2$.
- (A) 14
 - (B) 24
 - (C) 11
 - (D) 9
12. Melting of ice into water is a:
- (A) an explosion
 - (B) chemical change
 - (C) irreversible change
 - (D) physical change
13. How many lines of symmetry does an equilateral triangle have?
- (A) 0
 - (B) 3
 - (C) 1
 - (D) 2
14. When light reflects from a plane mirror, the angle of incidence is:
- (A) larger than the angle of reflection
 - (B) smaller than the angle of reflection
 - (C) equal to the angle of reflection
 - (D) always zero
15. Brackets in an arithmetic expression are an instruction to:
- (A) work out whatever is inside them first
 - (B) add the numbers inside to the answer twice
 - (C) ignore the numbers inside
 - (D) multiply the numbers inside

16. Rusting of an iron gate is a:
- (A) physical change
 - (B) chemical change
 - (C) a reversible change
 - (D) a change of state only
17. How many lines of symmetry does a circle have?
- (A) 1
 - (B) infinitely many
 - (C) 8
 - (D) 4
18. The image of your face in a flat (plane) mirror is:
- (A) smaller than the object
 - (B) formed on a screen behind the mirror
 - (C) virtual, erect, the same size, and laterally inverted
 - (D) real, upside down, and magnified
19. Evaluate $20 - 6 \div 2$.
- (A) 14
 - (B) 7
 - (C) 17
 - (D) 10
20. Cutting a sheet of paper into small pieces with scissors is a:
- (A) an irreversible chemical reaction
 - (B) burning
 - (C) physical change
 - (D) chemical change
21. An isosceles triangle (only two sides equal) has how many lines of symmetry?
- (A) 0
 - (B) 1
 - (C) 3
 - (D) 2
22. In a plane mirror, the image appears to be:
- (A) twice as far behind as the object is in front
 - (B) as far behind the mirror as the object is in front of it
 - (C) right on the surface of the mirror
 - (D) at the centre of the room
23. Evaluate $12 \div (6 - 2)$.
- (A) 6
 - (B) 3
 - (C) 0
 - (D) 4

24. Burning of paper is a:
- (A) a change of state
 - (B) a reversible change
 - (C) chemical change
 - (D) physical change
25. How many lines of symmetry does a scalene triangle (all sides different) have?
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
26. 'Lateral inversion' in a plane mirror means:
- (A) the image becomes smaller
 - (B) the left and right of the image are interchanged
 - (C) top and bottom are interchanged
 - (D) the colours are reversed
27. In the expression $5 + 7 - 3$, the separate terms are:
- (A) 5, +7 and -3
 - (B) 5, 7 and 3 with no signs
 - (C) just 5 and 7
 - (D) $5 \times 7 \times 3$
28. Most physical changes are:
- (A) changes that form new substances
 - (B) always irreversible
 - (C) always explosive
 - (D) reversible (they can usually be undone)
29. Which capital letter has a vertical line of symmetry?
- (A) P
 - (B) R
 - (C) A
 - (D) F
30. A mirror whose reflecting surface curves inward (like the inside of a spoon) is a:
- (A) concave mirror
 - (B) plane mirror
 - (C) transparent lens
 - (D) convex mirror
31. Which sign correctly fills the gap: $(4 + 5)$ ____ $(5 + 4)$?
- (A) < (less than)
 - (B) cannot be compared
 - (C) > (greater than)
 - (D) = (they are equal)

32. Most chemical changes are:
- (A) the same as a change of state
 - (B) usually permanent (hard to reverse)
 - (C) never accompanied by energy change
 - (D) always easy to reverse
33. Which capital letter has a HORIZONTAL line of symmetry?
- (A) F
 - (B) R
 - (C) A
 - (D) B
34. A mirror whose reflecting surface bulges outward (like the back of a spoon) is a:
- (A) magnifying lens
 - (B) convex mirror
 - (C) plane mirror
 - (D) concave mirror
35. Compare the two expressions: 7×3 ____ $7 + 3$.
- (A) they cannot be compared
 - (B) = (they are equal)
 - (C) > (the product is greater)
 - (D) < (the sum is greater)
36. For iron to rust, BOTH of these must be present:
- (A) only heat and light
 - (B) nitrogen and salt
 - (C) oxygen (air) and water (moisture)
 - (D) carbon dioxide and oil
37. How many lines of symmetry does the capital letter H have?
- (A) 0
 - (B) 4
 - (C) 2
 - (D) 1
38. A convex mirror always forms an image that is:
- (A) real and on a screen
 - (B) smaller (diminished) and erect
 - (C) the same size as the object
 - (D) larger and upside down
39. Evaluate $6 \times (10 + 2)$.
- (A) 72
 - (B) 38
 - (C) 62
 - (D) 18

40. The brown, flaky substance that forms on rusting iron is:
- (A) rust, which is iron oxide
 - (B) a kind of paint
 - (C) pure iron in powder form
 - (D) carbon (soot)
41. A regular pentagon has how many lines of symmetry?
- (A) 5
 - (B) 1
 - (C) 10
 - (D) 4
42. Convex mirrors are used as rear-view mirrors in vehicles mainly because they:
- (A) absorb the glare of headlights
 - (B) show an upside-down image of the road
 - (C) magnify approaching vehicles
 - (D) give a wider field of view
43. Using the spreading-out (distributive) idea, 8×13 equals $8 \times 10 + 8 \times 3 =$
- (A) 96
 - (B) 84
 - (C) 110
 - (D) 104
44. One common way to protect iron objects from rusting is to:
- (A) coat them with paint or a layer of another metal
 - (B) keep them in damp air
 - (C) sprinkle salt water on them
 - (D) leave bare iron in the rain
45. A regular hexagon has how many lines of symmetry?
- (A) 3
 - (B) 6
 - (C) 12
 - (D) 4
46. When the object is very close to it, a concave mirror can form an image that is:
- (A) always smaller and upright
 - (B) flat and colourless
 - (C) magnified (enlarged) and erect
 - (D) the same size as the object
47. If you add the SAME number to both sides of a true equality, the two sides:
- (A) stay equal to each other
 - (B) both become zero
 - (C) swap their values
 - (D) become unequal

48. Iron filings and sulphur powder simply stirred together form a mixture, but HEATING them strongly forms iron sulphide. The heating is a:

- (A) physical change
- (B) the same as mixing them cold
- (C) only a change of colour
- (D) chemical change

49. The ORDER of rotational symmetry of a square is:

- (A) 2
- (B) 1
- (C) 4
- (D) 8

50. A dentist uses a concave mirror because it:

- (A) makes the teeth look smaller
- (B) shows a wide view of the whole mouth
- (C) turns the teeth upside down
- (D) gives an enlarged image of the teeth

51. Given that $15 + 9 = 24$, what sign fills the gap: $(15 + 9 + 6) \text{ ____ } (24 + 6)$?

- (A) = (still equal)
- (B) > (left is greater)
- (C) < (left is smaller)
- (D) cannot be decided

52. Dissolving sugar in water is a:

- (A) chemical change
- (B) an irreversible change
- (C) burning
- (D) physical change

53. The smallest angle by which a square must be turned to look the same is:

- (A) 360 degrees
- (B) 90 degrees
- (C) 45 degrees
- (D) 180 degrees

54. Ordinary white light (sunlight) is actually made up of:

- (A) only black and white
- (B) a single pure colour
- (C) seven colours mixed together
- (D) only red and blue

55. Evaluate $2 \times 3 + 4 \times 5$.

- (A) 50
- (B) 14
- (C) 26
- (D) 70

56. The souring of milk into curd is a:

- (A) chemical change
- (B) physical change
- (C) a reversible change
- (D) a change of state only

57. The order of rotational symmetry of an equilateral triangle is:

- (A) 3
- (B) 6
- (C) 2
- (D) 1

58. The splitting of white light into its component colours is called:

- (A) dispersion
- (B) reflection
- (C) condensation
- (D) evaporation

59. Where should brackets go to make $3 + 5 \times 2$ equal 16?

- (A) $(3 + 5 \times 2)$
- (B) $(3 + 5) \times 2$
- (C) $3 \times (5 + 2)$
- (D) $3 + (5 \times 2)$

60. When carbon dioxide gas is passed through clear lime water, the lime water:

- (A) stays perfectly clear
- (B) freezes solid
- (C) turns bright blue
- (D) turns milky (a chemical change)

61. Through what smallest angle must an equilateral triangle be turned to look the same?

- (A) 90 degrees
- (B) 180 degrees
- (C) 60 degrees
- (D) 120 degrees

62. A rainbow forms in the sky because raindrops act like tiny:

- (A) filters that remove all colour
- (B) prisms that split sunlight into colours
- (C) lenses that magnify the Sun
- (D) mirrors that only reflect light

63. Evaluate $100 - (40 + 25)$.
- (A) 35
 - (B) 15
 - (C) 85
 - (D) 65
64. Photosynthesis, in which a leaf makes food from carbon dioxide and water, is a:
- (A) just the movement of water
 - (B) chemical change
 - (C) a change of state
 - (D) physical change
65. A figure that looks the same ONLY after a full 360 turn has rotational symmetry of order:
- (A) 4
 - (B) 0
 - (C) 2
 - (D) 1
66. Listed from one end of the spectrum to the other (VIBGYOR), the seven colours run from:
- (A) violet at one end to red at the other
 - (B) black at one end to white at the other
 - (C) brown at one end to grey at the other
 - (D) pink at one end to gold at the other
67. Taking 7 out as a common factor, $7 \times 6 - 7 \times 2$ equals $7 \times (6 - 2) =$
- (A) 28
 - (B) 56
 - (C) 12
 - (D) 32
68. Slowly evaporating a copper sulphate solution to obtain large, regular blue crystals is an example of:
- (A) a chemical change forming a new compound
 - (B) burning of the solution
 - (C) crystallisation, a physical change
 - (D) rusting of the copper
69. The order of rotational symmetry of a rectangle (not a square) is:
- (A) 3
 - (B) 1
 - (C) 4
 - (D) 2

70. A lens that is thicker in the middle than at the edges is a:
- (A) convex mirror
 - (B) concave (diverging) lens
 - (C) convex (converging) lens
 - (D) plane mirror
71. Which is larger: $12 + 12 + 12$ or 12×3 ?
- (A) $12 + 12 + 12$ is larger
 - (B) they are equal
 - (C) 12×3 is smaller
 - (D) 12×3 is larger
72. The hardening (setting) of cement after water is added is a:
- (A) chemical change
 - (B) a reversible change
 - (C) just drying like a puddle
 - (D) physical change
73. Which capital letter has rotational symmetry of order 2 (looks the same after a half turn)?
- (A) S
 - (B) A
 - (C) T
 - (D) B
74. A lens that is thinner in the middle than at the edges is a:
- (A) concave mirror
 - (B) concave (diverging) lens
 - (C) plane glass sheet
 - (D) convex (converging) lens
75. Removing the brackets, $9 + (4 + 6)$ equals:
- (A) 54
 - (B) 9
 - (C) 15
 - (D) 19
76. Cutting wood into pieces is a physical change, but burning wood is a:
- (A) physical change
 - (B) the same as cutting it
 - (C) a reversible change
 - (D) chemical change
77. A regular polygon with n equal sides has how many lines of symmetry?
- (A) 4
 - (B) $n - 1$
 - (C) n
 - (D) $2n$

- 78.** A convex lens can act as a magnifying glass because it forms a:
- (A) real image on the lens itself
 - (B) magnified, erect image of a nearby object
 - (C) smaller, inverted image
 - (D) colourless shadow of the object
- 79.** A candle 18 cm tall burns down 2 cm each hour. The expression for its height left after h hours is $18 - 2 \times h$. After 4 hours the height left is:
- (A) 72 cm
 - (B) 10 cm
 - (C) 16 cm
 - (D) 8 cm
- 80.** Coating iron with a thin layer of ZINC to protect it from rust is called:
- (A) crystallisation
 - (B) condensation
 - (C) evaporation
 - (D) galvanisation
- 81.** You hold the capital letter A in front of a vertical plane mirror and its reflection looks exactly like A. This is because A has:
- (A) a vertical line of symmetry
 - (B) rotational symmetry of order 2
 - (C) a horizontal line of symmetry
 - (D) no line of symmetry
- 82.** The picture thrown onto a cinema screen, where the light actually meets, is an example of a:
- (A) shadow
 - (B) mirror image
 - (C) virtual image
 - (D) real image
- 83.** Evaluate $50 - 2 \times (5 + 3)$.
- (A) 34
 - (B) 16
 - (C) 384
 - (D) 42
- 84.** Stainless steel resists rusting because it is an alloy of iron mixed mainly with:
- (A) rust and ash
 - (B) chromium (and some nickel)
 - (C) sugar and salt
 - (D) zinc and tin only

85. The line of symmetry of the capital letter T is:

- (A) diagonal
- (B) horizontal
- (C) both vertical and horizontal
- (D) vertical

86. The image in a plane mirror cannot be caught on a screen, so it is described as:

- (A) a shadow
- (B) a virtual image
- (C) a dispersed image
- (D) a real image

87. Evaluate $1000 / (10 \times 10)$.

- (A) 1000
- (B) 1
- (C) 10
- (D) 100

88. A 20 g iron nail gains mass as it rusts, because rust contains oxygen taken from the air. If it gains 3 g, the rusted nail's mass is:

- (A) 20 g
- (B) 23 g
- (C) 17 g
- (D) 60 g

89. A rhombus (a slanted square) has how many lines of symmetry?

- (A) 0
- (B) 4
- (C) 2
- (D) 1

90. The word AMBULANCE is printed reversed on the front of the vehicle so that, in a driver's mirror, it:

- (A) reads the correct way round
- (B) turns upside down
- (C) looks even more reversed
- (D) changes colour

91. The expression $3 \times n$ gives what value when $n = 7$?

- (A) 37
- (B) 10
- (C) 4
- (D) 21

92. Boiling water until it turns into steam is a:
- (A) an irreversible change
 - (B) physical change
 - (C) chemical change
 - (D) burning of the water
93. Through what smallest angle must a regular hexagon be turned to look the same?
- (A) 90 degrees
 - (B) 30 degrees
 - (C) 120 degrees
 - (D) 60 degrees
94. A green leaf looks green because, of all the colours in white light, it mostly:
- (A) turns all light into green
 - (B) creates green light of its own
 - (C) reflects green light and absorbs the rest
 - (D) absorbs green light and reflects the rest
95. Evaluate $4 + 4 / 4 + 4$.
- (A) 3
 - (B) 9
 - (C) 16
 - (D) 12
96. A burning candle shows BOTH kinds of change: the wax melting near the flame is physical, while the wax burning is:
- (A) a chemical change
 - (B) no change at all
 - (C) a change of state like melting
 - (D) also just a physical change
97. What is the order of rotational symmetry of a circle?
- (A) 360
 - (B) 1
 - (C) infinite
 - (D) 4
98. A real image differs from a virtual image because a real image:
- (A) is always upright
 - (B) appears behind a mirror
 - (C) can be obtained on a screen
 - (D) is always smaller

99. Without full calculation, comparing 25×4 and $25 + 4$, the product is:

- (A) impossible to compare
- (B) much smaller than the sum
- (C) much larger than the sum
- (D) exactly equal to the sum

100. Adding baking soda to lemon juice produces lots of fizzing bubbles of gas. This is a:

- (A) chemical change
- (B) a change of state only
- (C) simple dissolving
- (D) physical change